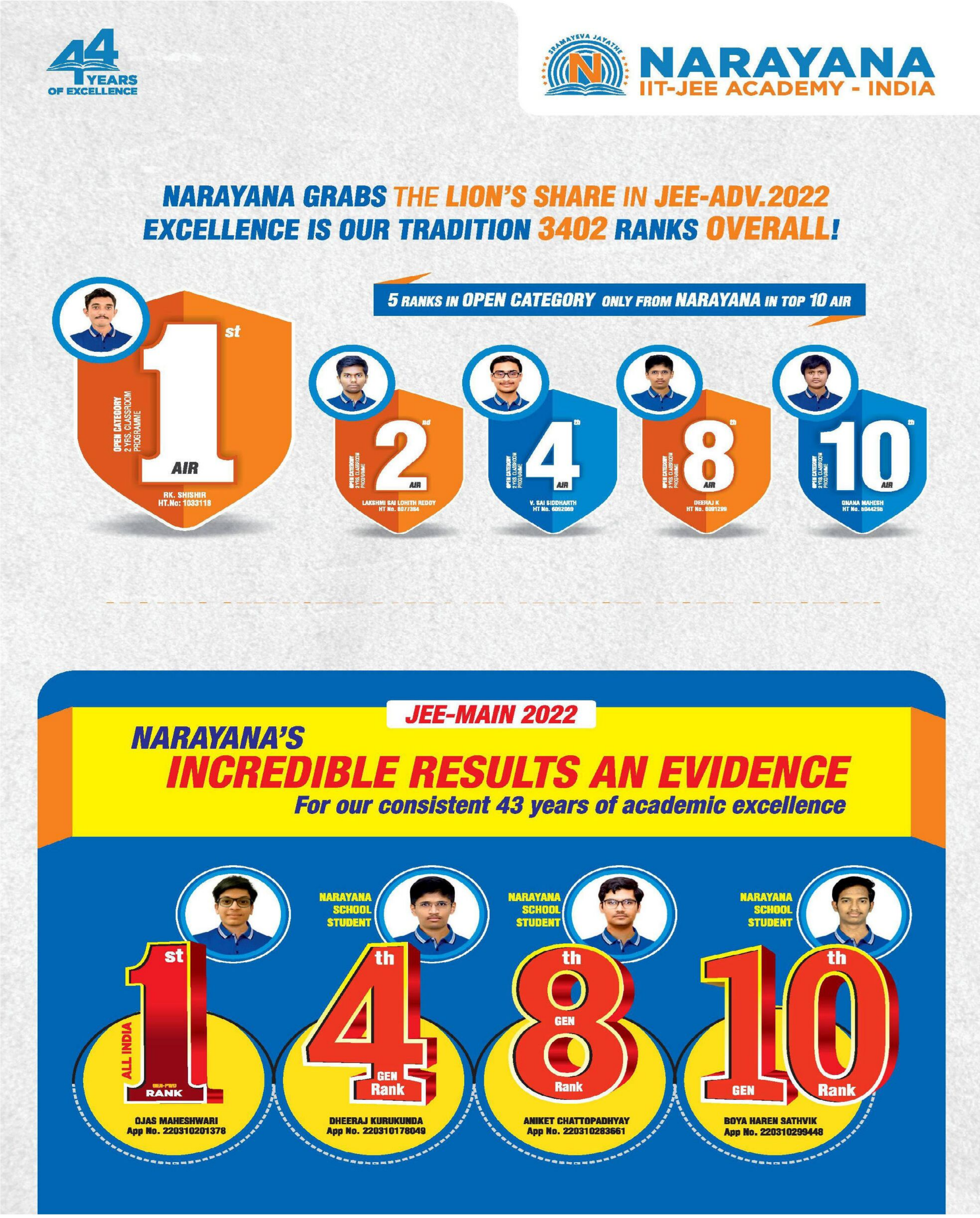


CHEMISTRY

INORGANIC CHEMISTRY PART-II(SR)

JEE MAIN IMPORTANT QUESTIONS

INORGANIC CHEMISTRY PART-II(SR)

COORDINATION COMPOUNDS

1. Geometry, hybridization, and magnetic moment of the ions $\left[Ni(CN)_4\right]^{2-}$, $[MnBr_4]^{2-}$ and $[FeF_6]^{3-}$, respectively are:

(A) Tetrahedral, square planar, octahedral: $sp^3, dsp^2,\ sp^3d^2 : 5.9, 0, 4.9$

(B) Tetrahedral, square planar, octahedral: $dsp^2, sp^3, sp^3d^2 : 0, 5.9, 4.9$

(C) Square planar, tetrahedral, octahedral: $dsp^2, sp^3, d^2sp^3 : 5.9, 4.9, 0$

(D) Square planar, tetrahedral, octahedral: $dsp^2, sp^3, sp^3d^2 : 0, 5.9, 5.9$

2. Which of the following statements is correct?

(A) With d^2sp^3 hybridization $\left[FeCl(CN)_4(O_2)\right]^{4-}$ complex is diamagnetic

(B) $[NiCl_4]^{2-}$ complex is more stable than $\left[Ni(dmg)_2\right]$ due to higher C.F.S.E. value

(C) $\left[V(CO)_6\right]$ is not very stable and easily reduces to $\left[V(CO)_6\right]^-$

(D) Ligands such as CO, CN^-, NO^+ are πe^- donor due to the presence of filled π-molecular orbital

3. Which of the following gives the maximum number of isomers?

(A) $\left[Co(NH_3)_4Cl_2\right]$ (B) $\left[Ni(en)(NH_3)_4\right]^{2+}$

(C) $\left[Ni(C_2O_4)(en)_2\right]$ (D) $\left[Cr(SCN)_2(NH_3)_4\right]^+$

4. Consider the following complex:

$\left[Co(NH_3)_5CO_3\right]ClO_4$

The coordination number, oxidation number, number of d-electrons, and number of unpaired d-electrons of the metal are respectively:

(A) 6, 3, 6, 0 (B) 7, 2, 7, 1 (C) 7, 1, 6, 4 (D) 6, 2, 7, 3

5. Select the correct order of magnetic moment (in B.M.) from the following options:

(A) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > \left[Fe(CN)_6\right]^{4-}$

(B) $\left[Fe(CN)_6\right]^{4-} > [MnCl_4]^{2-} > [CoCl_4]^{2-}$

(C) $\left[Fe(CN)_6\right]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

(D) $[MnCl_4]^{2-} > \left[Fe(CN)_6\right]^{4-} > [CoCl_4]^{2-}$

6. When $K_4\left[Fe(CN)_6\right]$ is treated with $FeCl_3$, a blue color is obtained. It is due to the formation of:

(A) $Fe^{II}\left[Fe^{III}(CN)_6\right]^-$
(B) $Fe^{III}\left[Fe^{II}(CN)_6\right]^-$
(C) Both (A) and (B)
(D) None of these

7. In the complex $\left[Pt(O_2)(en)_2(Br)\right]^{2+}$, nature of (O_2) is:

(A) Oxide ion
(B) Peroxide ion
(C) Superoxide ion
(D) Oxygen molecule

8. Among the properties (a) reducing, (b) oxidizing, (c) complexing, the set of properties shown by CN^- ion towards metal species is:

(A) a, b, c
(B) b, c
(C) c, a
(D) a, b

9. Select the correct statement for $\left[M(AB)b_2cd\right]$:

(A) All geometrical isomers are optically active
(B) It has four *trans* isomer with respect to b
(C) It has seven geometrical isomers
(D) It has three *cis* and two *trans* isomers with respect to b

10. The magnetic moment of a certain complex (A) of Co was found to be 4.89 B.M. and the EAN as 36. Co also forms complex (B) with magnetic moment 3.87 B.M. and EAN 37, and complex (C) with EAN as 36 but diamagnetic. Which of the following statements is true regarding the above observation?

(A) The oxidation states of Co in (A), (B), and (C) are +3, +2, and +3, respectively
(B) Complexes (A) and (B) have sp^3d^2 hybridization state while (C) has dsp^3 hybridization state
(C) The spin multiplicities of Co in (A), (B), and (C) are 4, 3, and 1, respectively
(D) The oxidation states of Co in (A), (B), and (C) are +6, +8, and +1, respectively

11. Find the number of total possible coordination isomers in $\left[Pt(NH_3)_4\right]\left[Cu(Cl)_4\right]$.

12. Find the total number of possible isomers of $\left[CrCl_3(CN)(NH_3)en\right]$

13. If $AgNO_3$ solution is added in excess to 1 M solution of $CoCl_3.XNH_3$, one mole of AgCl is formed. What is the value of x?

14. Find the number of t_{2g} and e_g electrons in $[NiF_6]^{2-}$.

Note: [If answer is 4 and 2 then represent as 42.]

15. Find the number of ligands which are non-classical ligand and π donor as well as π acceptor ligand.

$CO, PH_3, PF_3, C_3H_5^{\ominus}, C_5H_5^{\ominus}$

METALLARGY

16. Complexes formed in the following methods are:

(I) Mond's process for purification of nickel

(II) Removal of lead poisoning from the body

(III) Cyanide process for extraction of silver

(IV) Froth floatation process for separation of ZnS from galena ore by using depressant

	I	II	III	IV
(A)	$Ni(CO)_4$	$[Pb(EDTA)]^{2-}$	$[Ag(CN)_2]^{-}$	$[Zn(CN)_2]$
(B)	$Ni(CO)_4$	$[Pb(EDTA)]^{2-}$	$[Ag(CN)_2]^{-}$	$[Zn(CN)_4]^{2-}$
(C)	$Ni(CO)_6$	$[Pb(EDTA)]^{4-}$	$[Ag(CN)_2]^{-}$	$[Zn(CN)_6]^{4-}$
(D)	$Ni(CO)_4$	$[Pb(EDTA)]^{2-}$	$[Ag(CN)_4]^{3-}$	$[Zn(CN)_4]^{2-}$

17. Which of the following term is not related to A1-extraction?

(A) Serpek's process (B) Hall-Heroult process

(C) Thermite process (D) Hoop's process

18. The ignition mixture in "alumino thermite" process consists of a mixture of:

(A) Magnesium powder and BaO_2 or $KClO_3$

(B) Magnesium powder, aluminium, and $BaO_2 / KClO_3$

(C) Magnesium and aluminium powder

(D) Magnesium and aluminium oxides

19. In the electrolytic refining of silver, the anode mud obtained contains:

(A) Zn, Ag, and Au (B) Zu, Cu, Ag, and Au

(C) Au (D) Cu, Ag, and Au

20. Annealing of steel is the process of heating steel:

(A) To a bright red hot and then cooling it slowly

(B) To a bright red hot and then cooling it suddenly

(C) To a temperature much below redness and cooling it slowly

(D) None of the above

21. In the process of extraction of gold,

Roasted gold ore $+CN^{-}+H_2O \xrightarrow{O_2} [X]+OH^{-}$

$$[X]+Zn \rightarrow [Y]+Au$$

(A) $X=[Au(CN)_2]^{-}, Y=[Zn(CN)_4]^{2-}$

(B) $X=[Au(CN)_4]^{2-}, Y=[Zn(CN)_4]^{2-}$

(C) $X=[Au(CN)_2]^{-}, Y=[Zn(CN)_6]^{4-}$

(D) $X=[Au(CN)_4]^{-}, Y=[Zn(CN)_4]^{2-}$

22. Which of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly?

(A) CO_2 is more volatile than CS_2

(B) Metal sulphides are thermodynamically more stable than CS_2

(C) CO_2 is thermodynamically more stable than CS_2

(D) Metal sulphides are less stable than the corresponding oxides

23. Consider the following metallurgical processes:

1. Heating impure metal with CO and distilling the resulting volatile carbonyl (b.pt. $43^\circ C$) and finally decomposing at $150^\circ C$ to $230^\circ C$ to get the pure metal

2. Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged sulphide

3. Electrolyzing the molten electrolyte containing $CaCl_2$ to obtain the metal.

The processes used for obtaining sodium, nickel, and copper are, respectively,

(A) 1, 2, and 3 (B) 2, 3, and 1 (C) 3, 1, and 2 (D) 2, 1, and 3

24. Which of the following statement is correct?

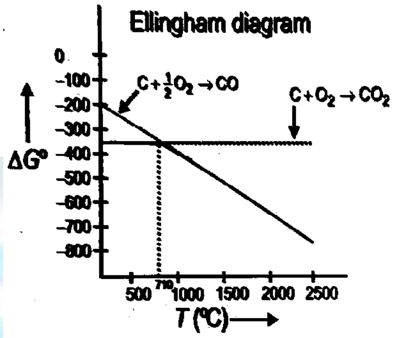
(A) Froth floatation method can only be used for sulphide ore

(B) Tin stone consists of wolframite as non-magnetic

(C) In cyanide process for the extraction of silver, Zn is used as leaching agent

(D) Bessemerization process is used in the extraction of copper from copper pyrite

25.



Which of the following is incorrect on the basis of the above Ellingham diagram for carbon?

(A) Up to $710^\circ C$, the reaction of formation of CO_2 is energetically more favorable, but above $710^\circ C$, the formation of CO is preferred

(B) In principle, carbon can be used to reduce any metal oxide at a sufficiently high temperature

(C) $\Delta S\left(C_{(s)} + 1/2 O_{2(g)} \rightarrow CO_{(g)}\right)$

$< \Delta S\left(C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}\right)$

(D) Carbon reduces many oxides at elevated temperature because ΔG° vs temperature line has a negative slope

26. How many metallic ores are concentrated by magnetic separation method from the given ores?

Cassiterite, Pyrolusite, Rutile, Magnetite, Galena, Cinnabar

27. Find the number of metals from the given metals which can be commercially purified by zone refining methods:

Si, Ge, Ga, Al, Ti, Zr

28. Find the number of metals which are commercially reduced by self-reduction from the given metals:

Fe, Al, Zn, Sn, Pb, Hg, Cu

29. How many metals are commercially extracted by

Al, Mg, Na, K, Ag, Hg, Ti, Th, Zr, B

30. How many reactions can show slag formation process from the given reactions?

(i) $SiO_2 + CaO \rightarrow CaSiO_3$

(ii) FeO + $SiO_2 \rightarrow FeSiO_3$

(iii) $CaO + P_2O_5 \rightarrow Ca_3(PO_4)_2$

(iv) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$

(v) $MgCO_3 + SiO_2 \rightarrow MgSiO_3 + CO_2$

QUALITATIVE INORGANIC CHEMISTRY

31. $MgSO_4$ on reaction with NH_4OH and Na_2HPO_4 forms a white crystalline precipitate. What is its formula?

(A) $Mg(NH_4)PO_4$ (B) $Mg_3(PO_4)12$

(C) $MgCl_2.MgSO_4$ (D) $MgSO_4$

32. Three test tubes A, B, C contain Pb^{2+}, Hg_2^{2+}, and Ag^+ (but unknown). To each, aqueous solution NaOH is added in excess. Following changes occur.

A: Black ppt. B: Brown ppt

C: White ppt. but dissolves in excess of NaOH A, B, and C contain, respectively:

(A) Pb^{2+}, Hg_2^{2+}, Ag^+ (B) Hg_2^{2+}, Ag^+, Pb^{2+},

(C) Ag^+, Pb^{2+}, Hg_2^{2+}, (D) Ag^+, Hg_2^{2+}, Pb^{2+},

33. If HCl is not added before passing H_2S in the second group, it may result in the:

(A) Incomplete precipitation of second group sulphide

(B) Precipitation of sulphides of cations belonging to subsequent groups

(C) Precipitation of sulphur

(D) Precipitation of lead as lead sulphide

34. Which metal gives blue ash when its salt is heated with Na_2CO_3 solid and $Co(NO_3)_2$ on a charcoal piece?

(A) Cu (B) Mg (C) Al (D) Zn

35. To avoid the precipitation of hydroxides of $Ni^{2+}, Co^{2+}, Zn^{2+}$, and Mn^{2+} along with those of Fe^{3+}, Al^{3+}, and Cr^{3+} the third group solution should be:

(A) Heated with a few drops of conc. HNO_3

(B) Treated with excess of NH_4Cl

(C) Concentrated

(D) None of these

36. In the second group of qualitative analysis, H_2S is passed through a solution acidified with HCl in order to:

(A) Limit the concentration of S^{2-} ions

(B) Increase the solubility of H_2S

(C) Increase the concentration of S^{2-} ions

(D) Add the Cl^- ions

37. A white colored salt forms white sublimate in dry heating test and also gives ammonia on heating with caustic soda solution. Which of the following test will be shown positive by the salt?

(A) It will give greenish yellow gas on reaction with conc. H_2SO_4

(B) It will give red gas by heating with $K_2Cr_2O_7$ and conc. H_2SO_4

(C) It will give colorless gas with dil. H_2SO_4

(D) It will give ring test

38. A hydrated metallic salt (A), light green in color, gives a white anhydrous residue (B) after being heated gradually. (B) is soluble in water and its aqueous solution reacts with NO to give a dark brown compound (C). (B) on strong heating gives a brown residue (D) and a mixture of two gases (E) and (F). The gaseous mixture, when passed through acidified permanganate, discharges the pink color:

(A) $ZnSO_4.7H_2O$ (B) $MgSO_4.7H_2O$ (C) $FeSO_4.7H_2O$ (D) $CuSO_4.5H_2O$

39. An inorganic Lewis acid (X) fumes in moist air, and intensity of fumes increases when a rod dipped in NH_4OH is brought near to it. An acidic solution of (X) on addition of NH_4Cl and NH_4OH gives a precipitate which dissolves in NaOH solution. An acidic solution of (X) does not give precipitate with H_2S. Hence, the compound (X) is:

(A) $FeCl_3$ (B) $AlCl_3$ (C) $SnCl_2$ (D) $ZnCl_2$

40. A colorless salt (X) is soluble in water and also in alcohol and amines. On strong heating, (X) gives a brown gas (Y) and a grey residue. (X) dissolves in ammonia to give a solution (Z)

which gives silver mirror with aldehydes. A solution of (X) is easily reduced by iron (II) sulphate. A solution of (X) also gives a brick red precipitate with potassium dichromate solution. Hence, choose the correct alternative:

	X	Y	Z
(A)	$Pb(NO_3)_2$	NO_2	Ag_2O
(B)	$AgNO_3$	NO	$[Ag(NH_3)_2]^+$
(C)	$AgNO_3$	NO_2	Ag_2O
(D)	$AgNO_3$	NO_2	$[Ag(NH_3)_2]^+$

41. Find the number of compounds which have yellow color ppt. from the given compounds:
$Ag_2CrO_4, pbCrO_4, Hg_2CrO_4, BaCrO_4$

42. $Na_2SO_3, NaCl, Na_2C_2O_4, Na_2HPO_4, Na_2CrO_4, NaNO_2, CH_3CO_2Na$ are separately treated with $AgNO_3$ solution. In how many cases is/are white ppt. obtained?

43. How many compounds liberate NH_3 on heating from the following?
$(NH_4)_2SO_4$, $(NH_4)_2CO_3$, NH_4Cl, NH_4NO_3, $(NH_4)_2Cr_2O_7$

44. How many water molecule(s) is/are present in compound which is used in borax bead test?

45. $Fe^{2+}(aq) + NO_3^-(aq) + H_2SO_4(conc) \rightarrow \text{Brown ring}$. The oxidation number of iron in brown ring complex is:

P-BLOCK ELEMENTS

46. $AlCl_3$ on hydrolysis gives:
(A) $Al_2O_3.H_2O$ (B) $Al(OH)_3$ (C) Al_2O_3 (D) $AlCl_3.6H_2O$

47. The correct order of thermal stability of hydrogen halides (H-X) is:
(A) HI > HBr > HCl > HF (B) HF > HCl > HBr > HI
(C) HCl > HF > HBr > HI (D) HI > HCl > HF > HBr

48. Hydrogen bond is strongest in:
(A) F – H – O (B) F – H – N
(C) F – H – F (D) All are equally strong

49. Which of the following is the correct order of strength of H-bonding in the given compound?
(A) $HF < NH_3$ (B) $H_2O > H_2O_2$
(C) $H_2O_2 > H_2O$ (D) $NH_3 > H_2O$

50. Laborer's working with phosphorus suffer from a disease in which bones decay. It is known as:
(A) Arthritis (B) Phossy jaw
(C) Rickets (D) Cancer

51. Pure phosphine is not combustible while impure phosphine is combustible: this combustibility is due to the presence of:

(A) P_2H_4 (B) N_2 (C) PH_5 (D) P_2O_5

52. P_4O_{10} on reacting with water does not form:

(A) Tetra metaphosphoric acid (B) Phosphorous acid

(C) Orthophosphoric acid (D) Pyrophosphoric acid

53. The color imparted by Co(II) compounds to glass is:

(A) Deep blue (B) Green (C) Yellow (D) Red

54. When PbO_2 reacts with conc. HNO_3, the gas evolved is:

(A) NO_2 (B) O_2 (C) N_2 (D) N_2O

55. By the action of conc. H_2SO_4, phosphorus changes to:

(A) Phosphorus acid (B) Metaphosphoric acid

(C) Orthophosphoric acid (D) Pyrophosphoric acid

56. Which of the following compound is responsible for catching fire spontaneously in Holme's singal?

(A) P_2H_4 (B) PH_3 (C) C_2H_4 (D) All of these

57. Which of the following statement regarding sulphur is incorrect?

(A) At $600^\circ C$, the gas mainly consists of S_2 molecules

(B) The oxidation state of sulphur is never less than +4 in its compounds

(C) S_2 molecule is paramagnetic

(D) The vapor at $200^\circ C$ consists mostly of S_8 rings

58. The liquefied metal expanding on solidification is:

(A) Al (B) Ga (C) Zn (D) Cu

59. A mixture of boron trichloride and hydrogen is subjected to silent electric discharge to form 'A' and HCl. 'A' is mixed with NH_3 and heated to $200^\circ C$ to form 'B'. The formula of 'B' is:

(A) H_3BO_3 (B) B_2O_3 (C) B_2H_6 (D) $B_3N_3H_6$

60. Which of the following is the correct order of increasing enthalpy of vaporization?

(A) $NH_3 < PH_3 < AsH_3$ (B) $AsH_3 < PH_3 < NH_3$

(C) $PH_3 < AsH_3 < NH_3$ (D) $NH_3 < AsH_3 < PH_3$

61. Chlorine reacts with excess of ammonia to form:

(A) NH_4Cl (B) $N_2 + HCl$ (C) $N_2 + NH_4Cl$ (D) $N_2 + NCl_3$

62. Aqueous solution of $Na_2S_2O_3$ in reaction with Cl_2 gives:

(A) $Na_2S_4O_6$ (B) $NaHSO_4$ (C) NaCl (D) NaOH

63. Which of the following compound have X – O – X linkage where 'X' is the so called central atom like P,S etc?

(A) $P_2O_8^{4-}$ (B) $S_2O_3^{2-}$ (C) $\gamma - SO_3$ (D) $S_2O_5^{2-}$

64. Which one of the following methyl diboranes does not exist?

(A) $B_2H_4(CH_3)_2$ (B) $B_2H_3(CH_3)_3$

(C) $B_2H_2(CH_3)_4$ (D) $B_2H(CH_3)_5$

65. Which of the following halogen oxides is ionic?

(A) I_4O_9 (B) I_2O_5 (C) BrO_2 (D) ClO_3

66. Antichlor is a compound:

(A) which absorbs chlorine

(B) Which removes excess of Cl_2 from a material

(C) Which liberates Cl_2 from bleaching powder

(D) Which acts as a catalyst in the manufacture of Cl_2

67. Peroxoborate in aqueous solution provides:

(A) $[B(OH)_4]^- + H_2O_2$ (B) $Na_2B_4O_7.10H_2O$

(C) HBO_2 (D) $Na[B_3O_3(OH)_4]$

68. Which reaction cannot give anhydrous $AlCl_3$?

(A) Heating of $AlCl_3.6H_2O$

(B) Passing dry HCl over heated aluminium powder

(C) Passing dry Cl_2 over heated aluminium powder

(D) Heating a mixture of alumina and coke in a current of dry Cl_2

69. The soldiers of Napolean army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to:

(A) An interaction with water vapor contained in humid air

(B) A change in crystalline structure of tin

(C) A change in the partial pressure of O_2 in air

(D) An interaction with N_2 of air at low temperature

70. Hydrolysis of one mole of peroxodisulphuric produces:

(A) Two moles of sulphuric acid

(B) Two moles of peroxomonosulphuric acid

(C) One mole of sulphuric acid and one mole of peroxomonosulphuric acid

(D) One mole each of sulphuric acid, peroxomonosulphuric acid and hydrogen peroxide

71. The product of oxidation of I^- with MnO_4^- in alkaline medium is:

(A) IO_3^- (B) I_2 (C) IO^- (D) IO_4^-

72. In silica, SiO_2, each silicon atom is bonded to how many oxygen atom?

73. The number of S-O-S bonds in sulphur trioxide trimer S_3O_9 is __________.

74. Find the number of ions having S-S bond from following:

$S_2O_4^{2-}, S_2O_6^{2-}, S_2O_2^{2-}, S_2O_3^{2-}, S_2O_7^{2-}$

75. Find the sum of the number of $P=O, P-O-P$ linkages in $P_5O_{16}{}^{-7}$ ion.

76. The atomicity of phosphorus is X and the P-P-P bond angle is Y. What are X and Y?

[If answer is 2 and 90° , represent as 290]

d-Block Elements

77. Which of the following form an alloy?

(A) Zn + Pb (B) Fe + Hg (C) Pt + Hg (D) Fe + Cr

78. For Ni and Pt different I.P. in kJ mol^{-1} are given below:

	$(IP)_1+(IP)_2$	$(IP)_3+(IP)_4$
Ni	2.49	8.80
Pt	2.60	6.70

Hence:

(A) Nickel (II) compounds tend to be thermodynamically more stable than platinum(II)

(B) Platinum(IV) compounds tend to be more stable than nickel(IV)

(C) Both are correct

(D) None is correct

79. $(NH_4)_2Cr_2O_7$ on heating gives a gas which is also given by:

(A) Heating NH_4NO_2 (B) Heating NH_4NO_3

(C) $Mg_3N_2 + H_2O$ (D) $Na + H_2O_2$

80. Arrange the following in order of their increasing thermal conductivity.

(A) Al, Ag, Cu (B) Cu, Ag, Al (C) Ag, Cu, Al (D) Al, Cu, Ag

81. When KI (excess) is added to:

(i) $CuSO_4$ (ii) $HgCl_2$ (iii) $Pb(NO_3)_2$

(A) A white ppt. of CuI in (i), an orange ppt. of HgI_2 in (ii), and a yellow ppt. of PbI_2 in (iii)

(B) A white ppt. of Cu_2I_2 in (i), a red ppt. dissolving to HgI_4^{2-} in (ii), and a yellow ppt. of PbI_2 in (iii)

(C) A white ppt. of CuI, HgI_2, and PbI_2 in each case

(D) None is correct

82. What are the species A and B in the following:

$$CrO_3 + H_2O \rightarrow A \xrightarrow{OH^-} B$$

(A) $H_2CrO_4, H_2Cr_2O_7$ (B) $H_2Cr_2O_7, Cr_2O_3$

(C) $CrO_4^{2-}, Cr_2O_7^{2-}$ (D) $H_2Cr_2O_7, CrO_4^{2-}$

83. $FeCr_2O_4 + Na_2CO_3 + O_2 \xrightarrow{\text{Fusion}} [X] \xrightarrow[H_2O]{H^+} [Y] \xrightarrow[H_2O_2]{H^+} [Z]$ Which of the following statements is true for the compounds [X], [Y], and [Z]?

(A) In all three compounds, the chromium is in +6 oxidation state

(B) [Z] is a deep blue-violet colored compound which decomposes rapidly in aqueous solution into Cr^{3+} and dioxygen

(C) Saturated solution of [Y] given bright orange compound, chromic anhydride, with concentrated H_2SO_4

(D) All of these

84. The atomic numbers of V, Cr, Mn, and Fe are respectively 23, 24, 25, and 26. Which one of these may be expected to have the highest second ionization enthalpy?

(A) Cr (B) Mn (C) Fe (D) V

85. Which of the following compounds will not give positive chromyl chloride test?

(A) $CuCl_2$ (B) $HgCl_2$ (C) $ZnCl_2$ (D) $C_6H_5\overset{+}{N}H_3Cl^-$

86. $FeCr_2O_4$ (chromite) is converted to Cr by following steps:

Chromite $\xrightarrow{I}$ Na_2CrO_4 $\xrightarrow{II}$ Cr_2O_3 $\xrightarrow{III}$ Cr I, II, and III are:

	I	II	III
(A)	Na_2CO_3 / air, Δ	C	C
(B)	Na_2CO_3 / air, Δ	C, Δ	Al, Δ
(C)	NaOH / air, Δ	C, Δ	Mg, Δ
(D)	conc. H_2SO_4, Δ	NH_4Cl, Δ	C, Δ

87. Lithopone is

A) $ZnSO_4 + BaS$ B) $ZnS + CaSO_4$ C) $ZnS + BaSO_4$ D) $ZnSO_4 + CaS$

88. Emerald green colored species is

A) $CuCl_4^{2-}$ B) $\left[Cu(H_2O)_4\right]^{2+}$ C) $CuCl_2.2H_2O$ D) $CuCl_2$ (anhydrous)

89. Which of the following can be reduced by ferrous ions?

A) Acidified $Cr_2O_7^{2-}$ B) $HgCl_2$

C) Acidified MnO_4^{2-} D) All of these

90. Which of the following is prepared using Kipp's apparatus waste?

A) $ZnSO_4$ B) $FeSO_4$ C) $CuSO_4$ D) $MnSO_4$

91. Which of the following has an electron is *(n – 1)d* subshell?

A) Ce, Gd and Eu B) Ce, Gd and Lu C) Gd, Tb and Eu D) Gd, Tb and Lu

F-BLOCK ELEMENTS

92. Consider the following statements in respect of lanthanoids:

(i) The basic strength of hydroxides of lanthanoids increases from $La(OH)_3$ to $Lu(OH)_3$.

(ii) The lanthanoid ions Lu^{3+}, Yb^{2+}, and Ce^{4+} are diamagnetic. Which of the statement(s) given above is/are correct?

(A) (i) Only (B) (ii) Only

(C) Both (i) and (ii) (D) Neither (i) nor (ii)

93. The actinoids showing +7 oxidation state are:

(A) U, Np (B) Pu, Am (C) Np, Pu (D) Am, Cm

94. Cerium (Z = 58) is an important member of the lanthanoids. Which of the following statement about cerium is incorrect?

(A) The common oxidation states of cerium are +3 and +4

(B) Cerium (IV) acts as an oxidizing agent

(C) The +4 oxidation state of cerium is more stable in solutions

(D) The +3 oxidation state of cerium is more stable than the +4 oxidation state

95. Which is not correct statement about the chemistry of 3d and 4f series elements

(A) 3d-elements show more oxidation states than 4f-series elements

(B) The energy difference between 3d and 4s orbitals is very little

(C) Europium(II) is more stable than Ce(II)

(D) The paramagnetic character in 3d-series elements increases from scandium to copper

96. In context of the lanthanoids, which of the following statement is not correct?

(A) There is a gradual decrease in the radii of the members with increasing atomic number in the series

(B) All the members exhibit +3 oxidation state

(C) Because of similar properties, the separation of lanthanoids is not easy

(D) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series

97. Which of the following is a synthetic radioactive lanthanoid?

A) Tm B) Sm C) Pm D) Pr

98. Naturally occurring actinoid are

A) Th, Pu and U B) Th, Pa and Am C) Pa, Cm and Cf D) Th, Pa and U

99. The three lanthanoids which exhibit +2 oxidation state are

A) Sm, Tb, Gd B) Sm, Eu, Yb C) La, Gd, Lu D) Yb, Pm, Sm

100. Pick the incorrect statement

A) Actinoids have higher tendency of complex as compared to lanthanoids

B) All actinoids are radioactive whereas all lanthanides except promethium are non-radioactive

C) Oxides and hydroxides of actinoids are more basic than that of lanthanoids

D) Highest oxidation state exhibited by actinoids is lower than that of lanthanoids

101. In aqueous solution, Eu^{2+} and Ce^{4+} acts as

A) oxidizing and reducing agent B) reducing and oxidizing agent

C) oxidizing agents D) reducing agent

ANSWER KEY

1	D	**2**	C	**3**	D	**4**	A	**5**	A
6	C	**7**	C	**8**	C	**9**	C	**10**	A
11	4	**12**	6	**13**	4	**14**	60	**15**	2
16	B	**17**	C	**18**	A	**19**	C	**20**	A
21	A	**22**	A	**23**	C	**24**	D	**25**	C
26	4	**27**	3	**28**	3	**29**	4	**30**	4
31	A	**32**	B	**33**	B	**34**	C	**35**	B
36	A	**37**	B	**38**	C	**39**	B	**40**	D
41	2	**42**	5	**43**	3	**44**	10	**45**	1
46	B	**47**	B	**48**	C	**49**	B	**50**	B
51	A	**52**	B	**53**	A	**54**	B	**55**	C
56	A	**57**	B	**58**	B	**59**	D	**60**	C
61	C	**62**	B	**63**	A	**64**	D	**65**	A
66	B	**67**	A	**68**	A	**69**	B	**70**	C
71	A	**72**	4	**73**	3	**74**	4	**75**	9
76	460	**77**	D	**78**	C	**79**	A	**80**	C
81	B	**82**	D	**83**	D	**84**	A	**85**	B
86	B	**87**	C	**88**	C	**89**	D	**90**	B
91	B	**92**	B	**93**	C	**94**	C	**95**	D
96	D	**97**	C	**98**	D	**99**	B	**100**	D
101	B								

COORDINATION COMPOUNDS
SOLUTIONS

5. $\left[MnCl_4\right]^{2-}$: $P > \Delta_t ; d^5$ - system $\mu_{eff} = 5.9$ B.M.

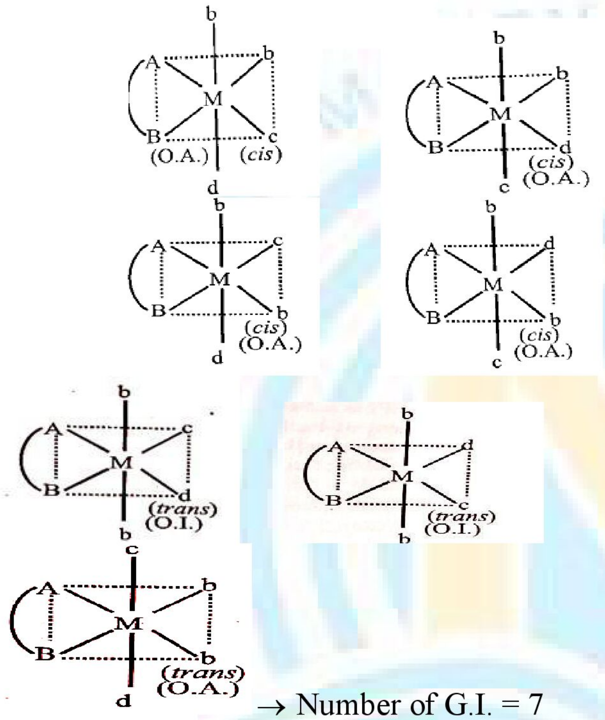
$\left[CoCl_4\right]^{2-}$: $P > \Delta_t ; d^7$ -system $\mu_{eff} = 3.89$ B.M

$\left[Fe(CN)_6\right]^{4-}$: $P < \Delta_0 ; d^6$ -system $\mu_{eff} = 0$ B.M.

6. X-ray studies have established that Prussian blue and Turnbull's blue are identical. Fe^{3+} partially oxidizes $\left[Fe^{II}(CN)_6\right]^{4-}$ forming some $\left[Fe^{III}(CN)_6\right]^{3-}$ Thus, $Fe\left[Fe(CN)_6\right]^{-}$ is formed.

8. $\overset{\ominus}{C}N$ acts as ligand and reducing agent.

9.



→ Number of G.I. = 7

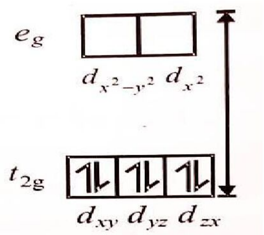
→ All cis : O. A. (Optically active)

→ One *trans* : O.A.

→ Two *trans* : O.I. (Optically inactive)

12. There are four geometrical isomers: two isomers of cis-dichloro and two isomers of trans-dichloro. All the two cis isomers are optically active. Hence, there are six isomers.

14.



Ni^{4+} in $\overset{+IV}{\left[NiF\right]_6^{2-}}$

METALLARGY SOLUTIONS

16. I Mond's porcess:

$$\underset{\text{Impure}}{Ni+4CO\uparrow}\xrightarrow{50^{\circ}C}\underset{\text{Volatile complex}}{\left[Ni(CO)_4\right]}\xrightarrow{230^{\circ}C}\underset{\text{pure}}{Ni+4CO\uparrow}$$

II Removal of unreacted AgBr from photographic plate:

$$AgBr+2Na_2S_2O_3 \rightleftharpoons Na_3\left[Ag(S_2O_3)_2\right]+NaBr$$

III Removal of lead poisoning:

$$Pb^{2+}+EDTA^{4-} \rightleftharpoons \left[Pb(EDTA)\right]^{2-}$$

IV Cyanide process:

$$2Ag+4NaCN+H_2O+\frac{1}{2}O_2 \rightleftharpoons \underset{+2\,NaOH\ \text{Native Ore}}{2Na\left[Ag(Cn)_2\right]}$$

V Separation of ZnS from PbS:

$$ZnS+CN^{-} \rightleftharpoons \left[Zn(CN)_4\right]^{2-}$$

17. Thermite process is related to Goldschmidt thermite reduction or welding where aluminium is used in powdered form (thermite) as reducing agent.

18. BaO_2 or $KClO_3$ provides oxygen to burn Mg.

22. The reduction process is on the thermodynamic stability of the products and not on their volatility.

QUALITATIVE INORGANIC SOLUTIONS

33. H_2S, although weak electrolyte, furnishes appreciable $\left[S^{2-}\right]$ to precipitate subsequent group radicals as sulphides.

34. $CoAlO_2$ is formed which is blue. Follow cobalt nitrate-charcoal test.

35. The presence of NH_4Cl in excess reduces degree of dissociation of NH_4OH to a considerable amount, and thus, $\left[OH^{-}\right]$ is just sufficient to cross over the ionic concentration product of hydroxides of III group elements to their K_{sp}.

36. The dissociation of H_2S is suppressed in the presence of HCl due to common ion effect, and thus, only II group radicals are precipitated.

39. (X)is $AlCl_3$

$$AlCl_3+\underset{\text{(moisture)}}{3H_2O}\rightarrow Al(OH)_3\downarrow+\underset{\text{(fumes)}}{3HCl}$$

$$AlCl_3+3NH_4OH\rightarrow Al(OH)_3\downarrow+3NH_4Cl$$

$$Al(OH)_3+NaOH\rightarrow NaAlO_2+2H_2O$$

$AlCl_3 + H_2S + H^+ \rightarrow \text{No ppt.}$

40. $\underset{(X)}{AgNO_3} \xrightarrow{\Delta} \underset{\text{(grey residue)}}{2Ag\downarrow} + \underset{(Y)}{2NO_2\uparrow} + O_2\uparrow$

$AgNO_3 + NH_4OH \rightarrow \left[Ag(NH_3)_2\right]^+$

$2\left[Ag(NH_3)_2\right]^+ + CH_3CHO + H_2O \rightarrow 2Ag\downarrow$

$+CH_3COO^- + 3NH_4^+ + NH_3$

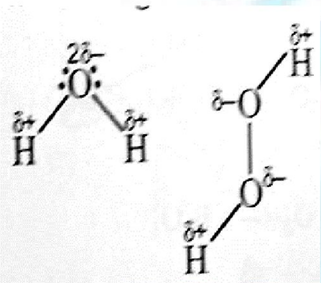
$Ag^+ + Fe^{2+} \rightarrow Ag\downarrow + Fe^{3+}$

$2AgNO_3 + K_2CrO_4 \rightarrow \underset{\text{(Brick red)}}{Ag_2CrO_4\downarrow} + 2KNO_3$

45. Oxidation number of iron is one $\left[Fe(H_2O)_5(NO)\right]SO_4$

p-Block Elements

49.



The strength of H-bonding is more in H_2O.

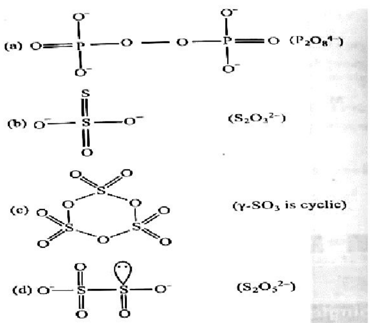
58. Ga is a soft silvery white metal and is liquid at room temperature. When it solidifies, it expands by 3.1 percent. Therefore, it should not be stored in glass or metal containers.

59. $2BCl_3 + 3H_2 \xrightarrow[\text{discharge}]{\text{electric}} B_2H_6 + 6HCl$

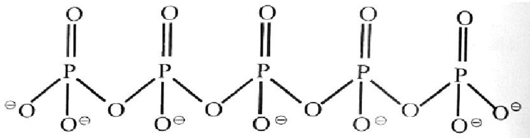
$B_2H_6 + 2NH_3 \rightarrow B_2H_6.2NH_3 \underset{B_3N_3H_6}{\downarrow 200^\circ C}$

62. $Na_2S_2O_3 + 4Cl_2 + 5H_2O \rightarrow 2NaHSO_4 + 8HCl$

63.



75.



No. of P = O bonds 5

No. of P – O – P bonds = 4

Total = 5 + 4 = 9

(d-Block Elements)

77. Iron and carbon form alloy which is called cementite (Fe_3C).

88. $CuCl_2.2H_2O$ is emerald green crystals

89. $Fe^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow Fe^{3+} + Cr^{3+} + H_2O$

$HgCl_2 + Fe^{2+} \rightarrow Hg_2Cl_2 + Fe^{3+}$

$Fe^{2+} + MnO_4^{2-} + H^+ \rightarrow Fe^{3+} + Mn^{2+} + H_2O$

90. Kipp's apparatus waste contains $FeSO_4$

$FeS + di.H_2SO_4 \rightarrow FeSO_4 + H_2S$

91. $Ce : 4f^1 5d^1 6s^2; Gd : 4f^7 5d^1 6s^2; Eu : 4f^7 6s^2; Tb : 4f^9 6s^2, Lu : 4f^{14} 5d^1 6s^2$

f-BLOCK ELEMENTS

92. $La(OH)_3$ is most basic in nature while $Lu(OH)_3$ least basic.

93. Np and PU actinoids showing +7 oxidation state.

94. The E° value for Ce^{+4} / Ce^{+3} is 1.74 V which suggests that is can oxidize water.

97. Promethium (Pm) is the only synthetic radioactive lanthanoid

98. All actinoids after uranium are man-made (transuranic element). Thus, Thorium (Th). Protactinium (Pa) and Uranium (U) are the three naturally occurring actinoids.

99. Sm, Eu, Tm and Yb all exhibit an oxidation state of +2 in addition to the most common +3 oxidation state.

100. Highest oxidation state of actinoids (+6) is greater than that of lanthanoids (+4).

101. Both Eu^{2+} and Ce^{4+} have to tendency to convert to the most common oxidation number of +3 as under:

$Eu^{2+}(\text{reducing}) \rightarrow Eu^{3+} + e^-$

$\underset{(\text{Oxidizing})}{Ce^{4+}} \xrightarrow{e^-} Ce^{3+}$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